LLM Web Dynamics: Tracing Model Collapse in a Network of LLMs

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Abstract

The increasing use of synthetic data from the public Internet has enhanced data usage efficiency in large language model (LLM) training. However, the potential threat of model collapse remains insufficiently explored. Existing studies primarily examine model collapse in a single model setting or rely solely on statistical surrogates. In this work, we introduce LLM Web Dynamics (LWD), an efficient framework for investigating model collapse at the network level. By simulating the Internet with a retrieval-augmented generation (RAG) database, we analyze the convergence pattern of model outputs. Furthermore, we provide theoretical guarantees for this convergence by drawing an analogy to interacting Gaussian Mixture Models.

1 Introduction

Synthetic data is crucial for training and scaling large language models. On the one hand, neural-scaling law assumes model performance will improve as the amount of training data increases (Kaplan et al., 2020); on the other hand, the pool of human-generated data is approaching its natural limit (Villalobos et al., 2024). Augmenting the training data with synthetic text has therefore become a common remedy (Man and Chahl, 2022). However, recent evidence suggested that the usage of synthetic data can lead to issues including parameter degradation (Dohmatob et al., 2024), amplified biases (Wyllie et al., 2024), and, in the most extreme cases, model collapse: a process characterized by the loss of information about the raw data distribution when an LLM is iteratively trained primarily on its own previous outputs (Shumailov et al., 2024). Analyzing the process of model collapse, and more generally the perturbations from synthetic data, is thus essential for LLM training.

In our work, we introduce the *LLM Web Dynamics* (LWD)—a framework that renders the full lifecycle of synthetic data *tractable*, *measurable*, and *theoretically valid*. Our contributions are threefold:

1. Network-level model-collapse metric. We design a network of interacting LLMs in which pretrained models from API communicate, receive opinions from each other, and update memory via retrieval-augmented generation (RAG). Individual responses are mapped into an embedding vector space and then converted to a pairwise distance matrix which quantitatively captures response similarities. As collapse unfolds and response similarity rises, this matrix shows drifting in geometry features accordingly, which provides a model-agnostic stability measuring metric. This metric reflects the stabilization of collective communicating dynamics of all participant LLMs.

2. Internet ecosystem simulator (API-only, cost-efficient). We anticipate a near-future feedback loop where synthetic texts generated by LLM agents populate the Internet and are later reused as training data or prompt augmentations in downstream applications (Martínez et al., 2023; Seddik et al., 2024). Our LWD framework constructs a virtual reality that mirrors real-world, AI-rich Internet environment by deploying a centralized RAG memory updating algorithm. Since each agent operates solely through API-accessible models, the framework approximates incremental fine-tuning without the need for retraining or additional fine-tuning, while also enabling the exploration of more complex scenarios.

3. Theoretical grounding via a Gaussian Mixture Model (GMM) proxy. To validate the collapse process of our LLM system analytically, we construct an analogue network of GMMs, motivated by Wang et al. (2025). For a fixed prompt, an LLM's answers form a probability distribution, and

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GMMs serve as density approximators (Goodfellow et al., 2016). This proxy offers theoretical guarantees for our claims and serves as a cost-efficient tool for gaining preliminary insights into LLM behavior at larger scales.

2 Related Work

This phenomenon of model degradation is well documented and has been observed across domains, tasks, and model types. Alemohammad et al. (2023) introduce self-consuming training loops and show that, without a constant infusion of fresh real data, models converge to "Model Autophagy Disorder" (MAD). In image generation, Bohacek and Farid (2025) demonstrate that even a small share of self-generated images used during retraining triggers self-poisoning, reducing both realism and diversity. Shumailov et al. (2024) first formalize this phenomenon as model collapse and demonstrates its existence in practice via GMMs, variational autoencoders (VAEs), and LLMs.

Quantifiable and interpretable model collapse have also attracted many attention in recent literature. Statistical surrogates are commonly used to provide instrumentally similar insights into this problem. In particular, regression-based methods are applied to simulate the iterative training of LLMs because they yield closed-form analytical risk expressions (Dohmatob et al., 2024; Gerstgrasser et al., 2024; Kazdan et al., 2025). Surrogates with stronger distribution assumptions including Gaussian distributions, Gaussian processes, and GMMs provide a theoretically interpretable and tightly controllable testbed whose generative nature closely mirrors that of LLMs (He et al., 2025; Suresh et al., 2024; Borji, 2024; Li et al., 2024; Alemohammad et al., 2023).

A crucial finding is that collapse dynamics shall be decisively influenced by the strategy for accumulating and infusing synthetic data across training iterations. A full replace paradigm (Schaeffer et al., 2025; Dey and Donoho, 2024) refers to entire updating in training data. Gerstgrasser et al. (2024) and Kazdan et al. (2025) validate that the accumulation of synthetic data from all previous iterations effectively prevents the collapse of the model. Bertrand et al. (2024) only keep the most recent batch of synthetic data and points out synthetic to real data ratio shows a more significant role in model collapse.

Recent work centers on establishing connection between model collapse and a varying synthetic-to-real ratio. Gillman et al. (2024) report that a moderate level of synthetic input helps preserve low-density diversity. He et al. (2025) prove that, for both mean and covariance estimation, the estimation error is minimized when the weight on real data equals the reciprocal of the golden ratio. Conversely, Dohmatob et al. (2024) show that collapse can arise even at very small amount of synthetic data, implying that the synthetic-to-real ratio must asymptotically approach zero to maintain model diversity. Similarly, Bertrand et al. (2024) argue that keeping the ratio below a certain threshold also promotes long-run stability of training.

These results highlight the *controllability* of model collapse. A systematic strategy for synthetic and real data combination shall lead to a framework of interpretable model uncertainty control, more efficient use of synthetic data, and, cheaper but safer scaling of LLMs. Despite these promises, current research still faces challenges. **(i) Disconnect from realistic web dynamics and synthetic data generation**. Most existing analyses rely on purely statistical models, not tailored to mimic real-world Internet ecosystems. Additionally, their experimental setups often involve collecting and recycling synthetic generations across simulated time, a process that lacks a clear real-world analogue. **(ii) Computational limitations.** Full scale LLM model collapse experiments require iterative retraining and therefore remain extremely expensive, constraining empirical validation.

Our LWD framework is designed to bridge these gaps. We describe our experimental design in Section 3, and present empirical results in Section 4, showing that the norm of the embedded distance matrix decreases during collapse in both LLM and GMM experiments. The limiting behavior of the GMM setup is proved in Appendix A. In Sections 5 and 6, we discuss our findings and outline directions for future work.

3 Methods and Experiments

3.1 Experiment: Synthetic Conversations on the Internet

In our LLM Web Dynamics (LWD) framework, n different LLM models discuss a certain topic and get updated by answers from other participants over time. Full fine-tuning at each time is infeasible and expensive, so each model instead refreshes its context via retrieval-augmented generation (RAG).

RAG is lightweight to deploy and has been proven repeatedly showing equal or better performance in particular tasks comparing to fine-tuning in empirical studies (Balaguer et al., 2024; Cheng et al., 2025; Singh et al., 2025; Gupta et al., 2024), making it a practical and low-cost surrogate for iterative learning and training within the framework of LWD.

Suppose we have n LLMs $\{\mathcal{M}_i\}_{i=1}^n$, each pretrained on different data. For instance, `GPT-3.5` is trained largely on English-dominant data (Brown et al., 2020), whereas `DeepSeek-67B` draws from a bilingual data of Chinese content (DeepSeek-AI et al., 2024). These geographic and linguistic differences give the models heterogeneous priors over vocabulary, style, and factual knowledge, which is essential for studying cross-model information propagation (AlKhamissi et al., 2024; Gupta et al., 2025). To measure model performance, we use a fixed query q and define a set of sentences containing information related to this query q. Intuitively, this set serves as the Internet which language models grab information from and contribute generated outputs to. In detail, at each time $t = 1, 2, \ldots$, each model augments itself by retrieving k_t sentences from this set to generate a response to the query. Here we allow each model to grab a different number of sentences (and hence different amounts of information) as t grows. In the standard RAG approach (Gao et al., 2024), LLM should retrieve the top k_t relevant sentences; in our case, the sentences in the set are all relevant to q by construction, so we draw k_t sentences uniformly at random from this set each time.

We denote this set of sentences at time t as $A^{(t)}$. If this set changes over time, each model's response would change accordingly, and thus we denote the response of model $i \in \{1, \ldots, n\}$ at time t to query q as $F_i^{(t)}(q)$. For a fixed query q, there is still randomness in an LLM's output, so $F_i^{(t)}(q)$ is a random variable whose distribution depends on the model at time t and the query. Our goal is to measure model performance, so at each t, we ask every model L iterations to get L responses $(f_i^{(t)}(q))_1, \cdots, (f_i^{(t)}(q))_L$ drawn independently from the distribution of $F_i^{(t)}(q)$.

At each time t, after all n models each generate L responses, we take one response from each model and "post" them to the "Internet" $A^{(t)}$. Under the i.i.d. assumption that we made previously, randomly picking one of L responses is probabilistically equivalent to simply taking the first one. Thus, we can define the set A at $t+1$ as:

$$A^{(t+1)} = A^{(t)} \cup \{(f_1^{(t)}(q))_1, \cdots, (f_n^{(t)}(q))_1\}.$$

Hence $|A^{(t+1)}| = |A^{(t)}| + n$, reflecting that the Internet is increasingly filled with synthetic answers. If we begin with $|A^{(0)}|$ human-generated sentences, then at time t, the probability of retrieving a synthetic sentence from $A^{(t)}$ is $\frac{nt}{|A^{(0)}|+nt}$ (since we choose to retrieve them uniformly at random), which increasingly approaches 1 as t increases. Recall that at each t, each model fetches k_t sentences from $A^{(t)}$. We want to use this experiment to discuss the case when LLMs get trained and updated by the information on the Internet, so it is reasonable to make k_t proportional to $|A^{(t)}|$, i.e., $k_t = \lfloor \beta \cdot |A^{(t)}| \rfloor$ for a fixed hyperparameter β.

Using this procedure (illustrated in Figure 1), we can observe how model responses evolve over time. To analyze the responses statistically, we use the open source embedding model `nomic-embed-v1.5` (Nussbaum et al., 2024), denoted as ϕ, which maps each text $(f_i^{(t)}(q))_l$ to a numeric vector $(\phi \circ f_i^{(t)}(q))_l \in \mathbb{R}^{768}, l = 1, \cdots, L$. This function ϕ allows us to quantify the similarity between two texts such that a smaller distance between embedding vectors indicates that the original two sentences are more semantically similar. Given these L vectors for each model i at time t, we can estimate the distribution of $(\phi \circ F_i^{(t)}(q))$. This can be challenging due to the high dimension of the embedding model, and one can adopt the framework of *Data-Kernel Perspective Space* (DKPS) (Helm et al., 2024), which reduces its dimension by Classical Multidimensional Scaling to get nice vector representations.

In addition to observing how the distribution evolves, we focus on estimating the distribution mean by averaging these L vectors to measure how model i responds to query q at time t:

$$X_i^{(t)} := \frac{1}{L} \sum_{l=1}^{L} \left(\phi \circ f_i^{(t)}(q)\right)_l \in \mathbb{R}^{768}. \quad (1)$$

We conjecture that as t increases, the distributions of $F_i^{(t)}(q)$ and $F_j^{(t)}(q)$ for $i \neq j$ increasingly resemble each other, which would then imply the same behavior for the sample means $X_i^{(t)}$ and $X_j^{(t)}$. In detail, we capture the pairwise model difference through a $n \times n$ matrix $D^{(t)}$ whose entries are de-